

Folding Calculus

Derivatives as Folds, Integrals as Unfolds, and Creases as Primary Objects

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Abstract

The fundamental operations of calculus have a simple geometric meaning: differentiation is **folding** (compressing global information to local slope), and integration is **unfolding** (expanding local slopes into global structure). At a 90-degree crease (e.g., $f(x) = |x|$ at $x=0$), folding reveals a set of possible slopes rather than a single number. This reframing makes creases central rather than exceptional and directly illuminates ReLU neural networks as layered folding machines. We formalize the framework, introduce crease density as a diagnostic, and present supporting experiments.

1. The 90-Degree Fold — The Core Intuition

(Insert image from PDF: title/cover page or Chapter 1 screenshot here)

Consider the graph of $f(x) = |x|$ — two straight lines meeting at a sharp 90-degree crease at $x=0$.

- **Folding at the crease:** Press the left and right arms flat together. All global shape information collapses to the local possibilities at the hinge: any supporting line with slope m where $-1 \leq m \leq 1$. This set *is* the subdifferential $\partial f(0) = [-1, 1]$.

- **Unfolding:** Take local slopes and integrate them. You recover piecewise linear functions consistent with the original crease.

This is exact mathematics. For smooth functions (no sharp crease, e.g. x^2), the fold yields a single tangent slope. The Fundamental Theorem of Calculus states that fold and unfold are inverses.

2. Formal Definitions

- **Fold (Differentiation):** Maps a function to its local linear approximations.

- **Unfold (Integration)**: Reconstructs the global function from local approximations.
- **Crease**: Point where the subdifferential is an interval (positive diameter) rather than a singleton.
- **Crease Density**: Fraction of ReLU pre-activations near zero in a neural network (quantifies undecided folds).

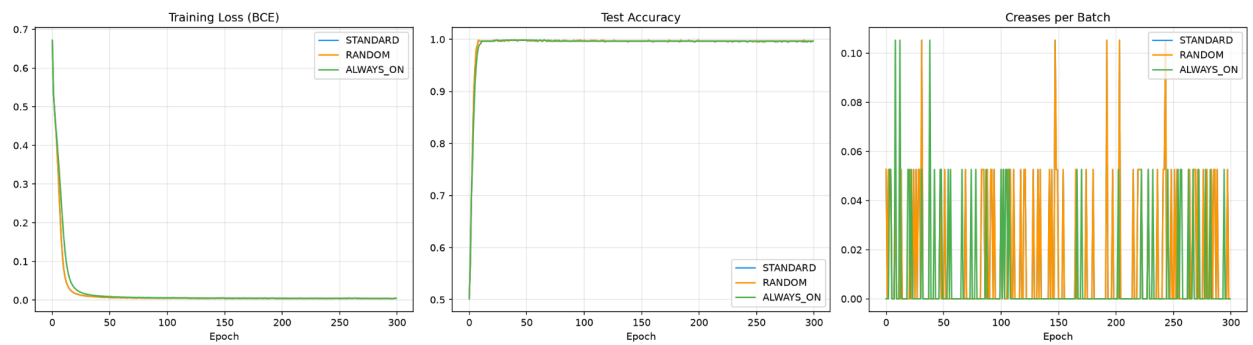
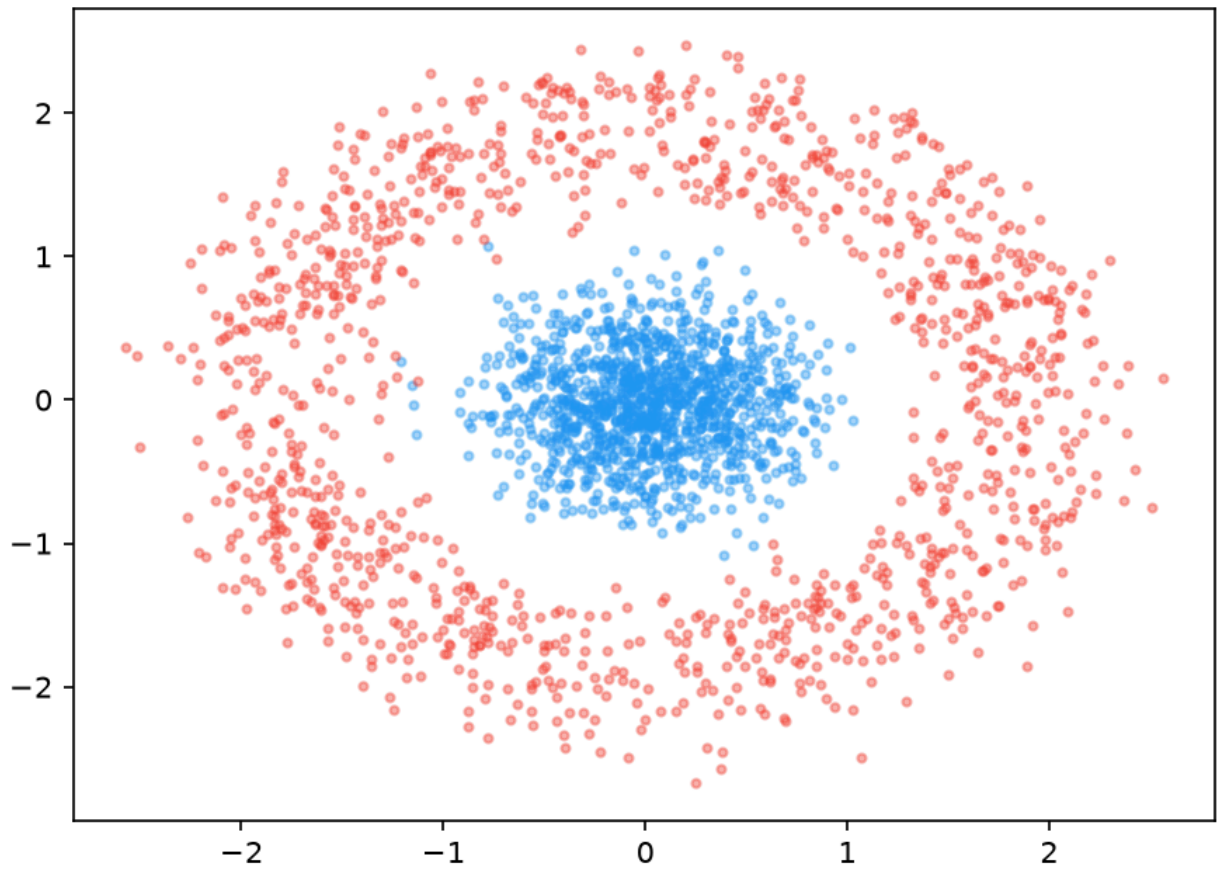
3. ReLU Networks as Layered Folding

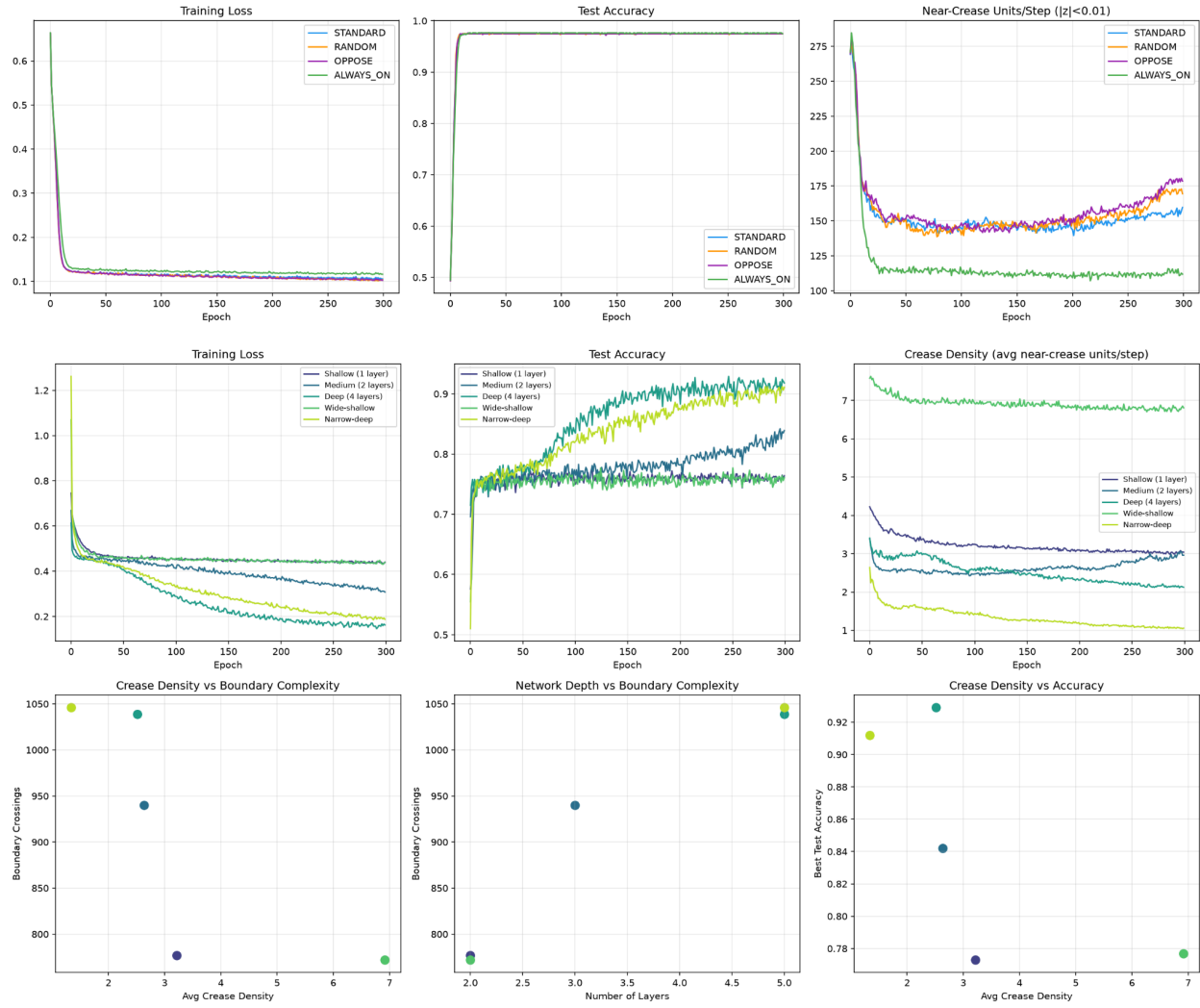
Each ReLU layer applies a 90-degree-style fold along a hyperplane. Deeper networks perform more folds, creating complex decision boundaries. Crease density tracks how many units remain at their switching hinges during training.

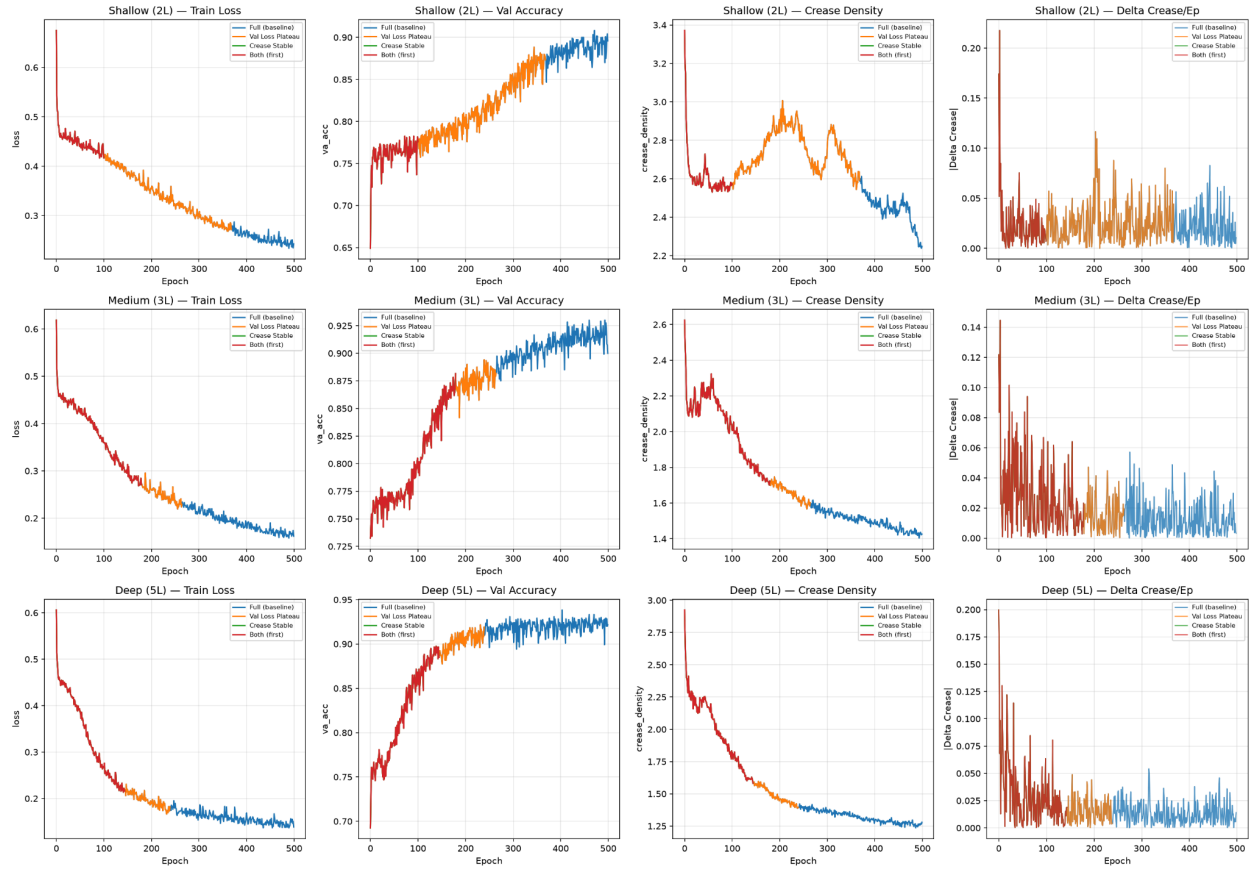


4. Experiments & Results

Ring Dataset (2D binary classification)







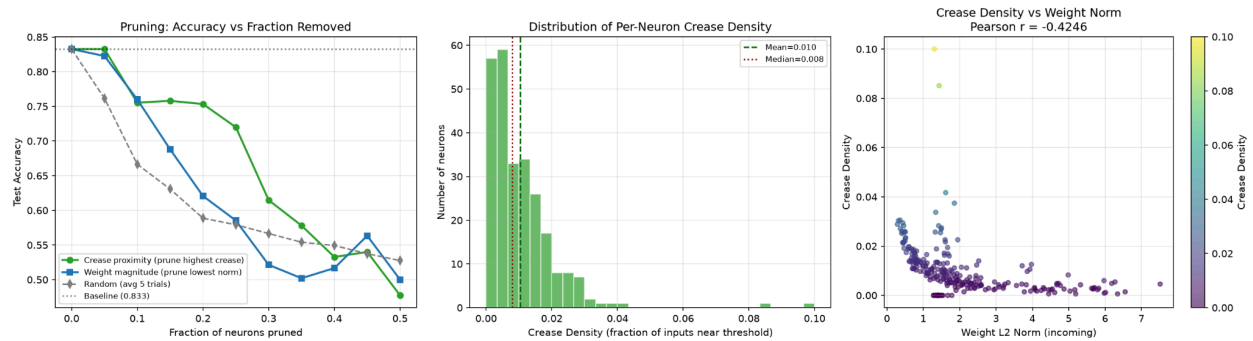
5. Discussion & Open Problems

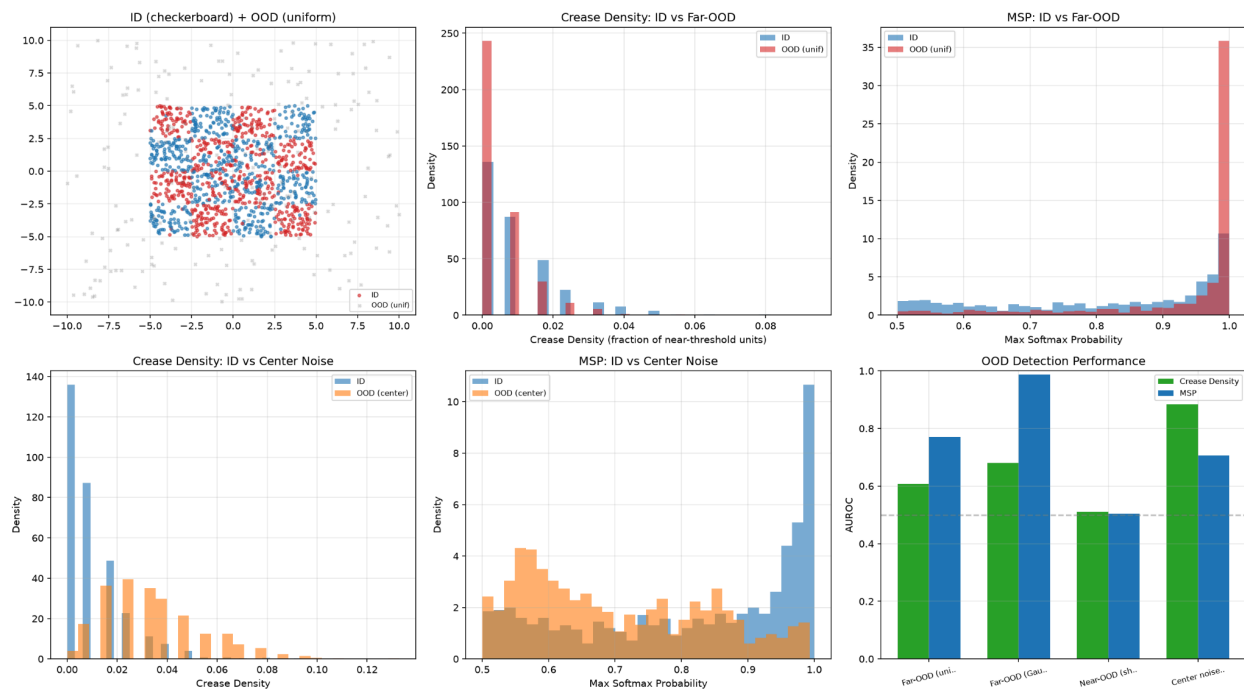
The 90-degree fold provides a unifying geometric language. Future work: formal fold depth complexity, soft creases for modern activations, and applications in geometry.

References

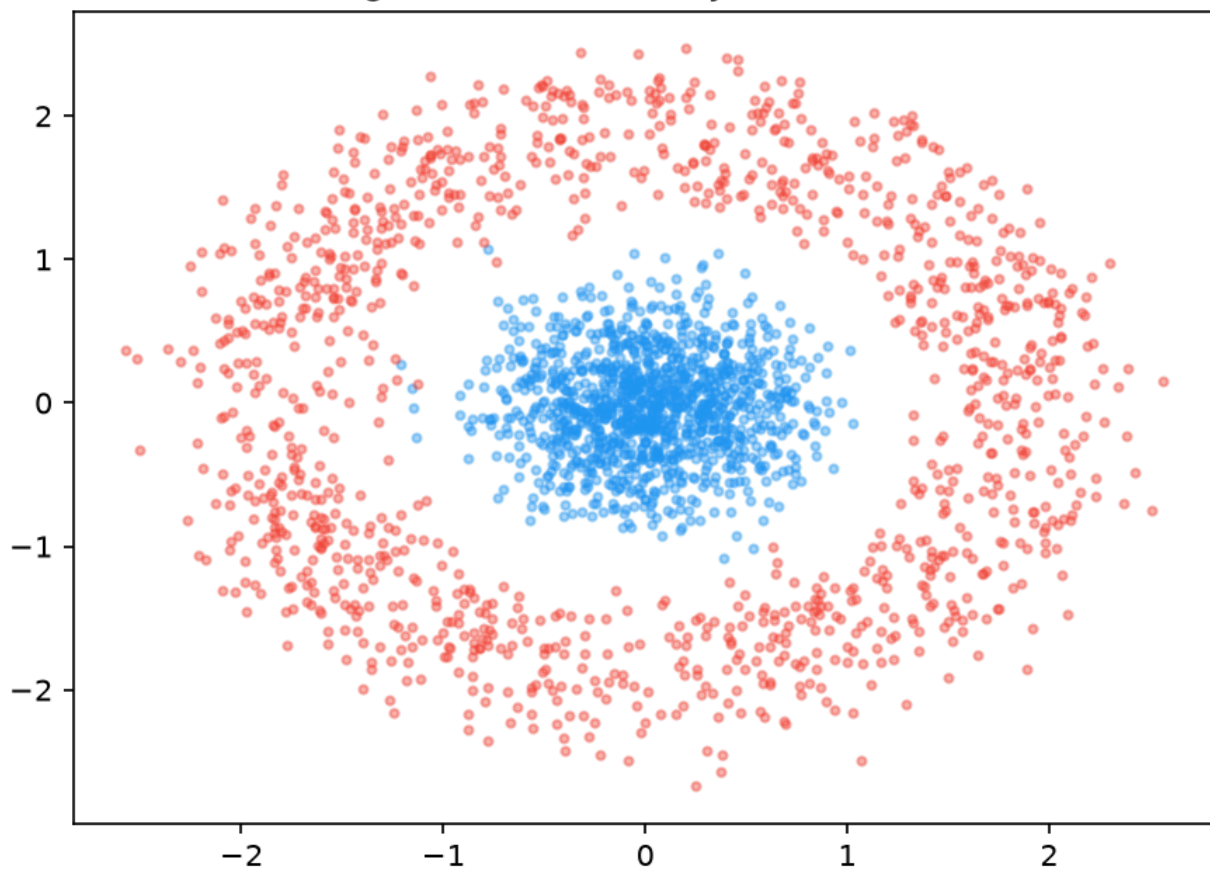
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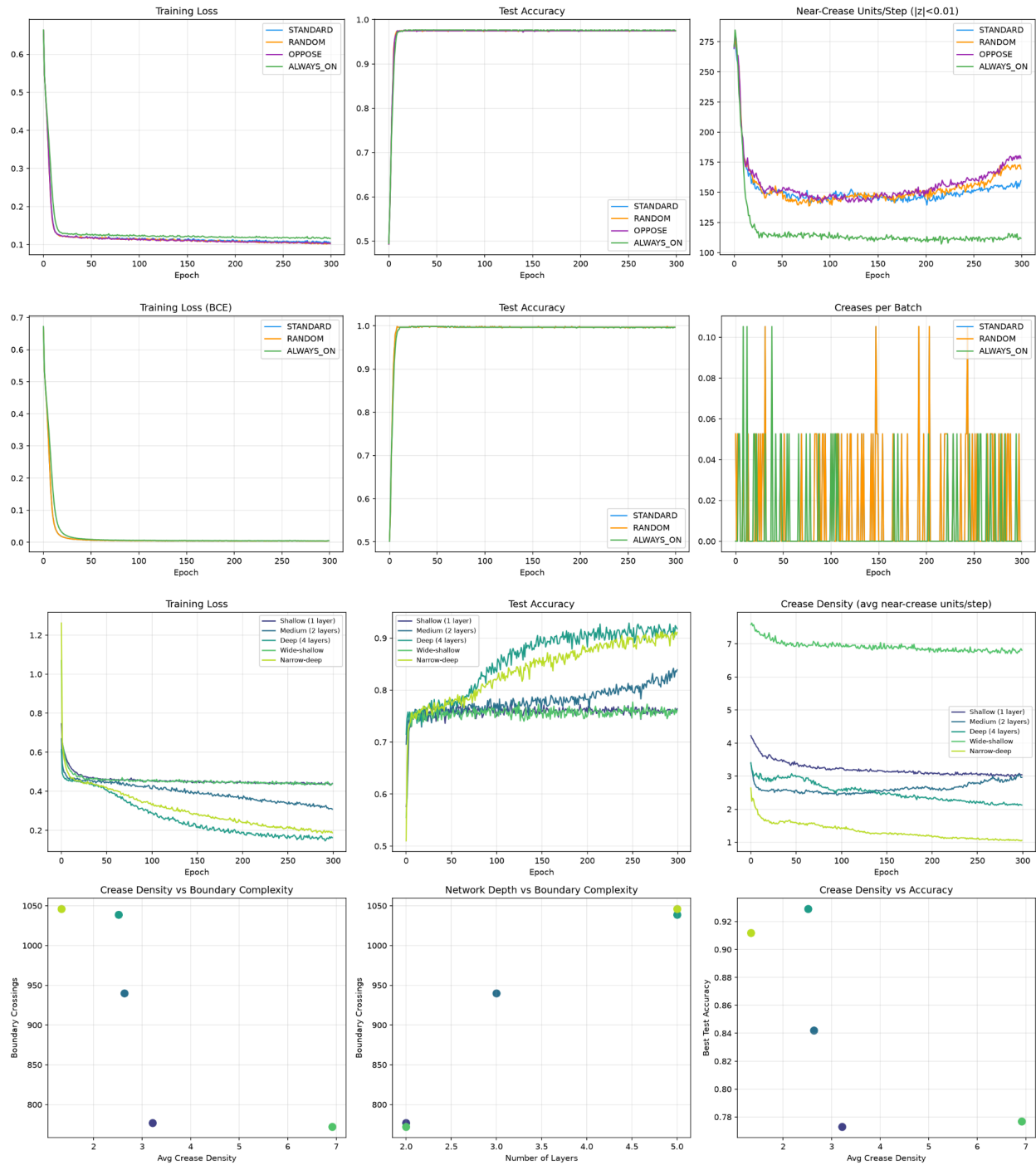
Appendix: Code & Visuals

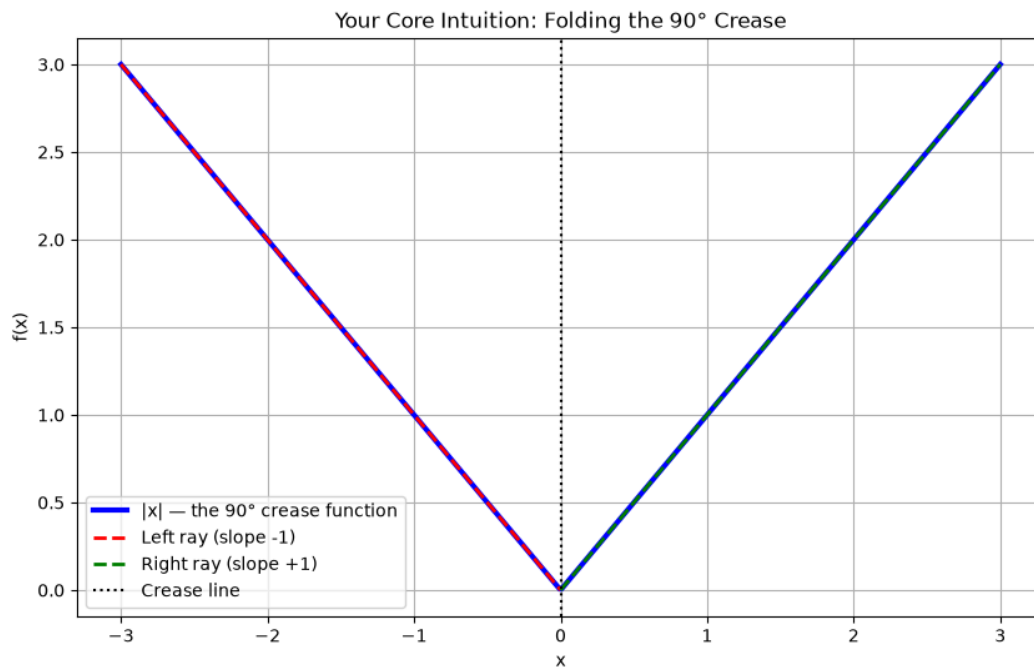
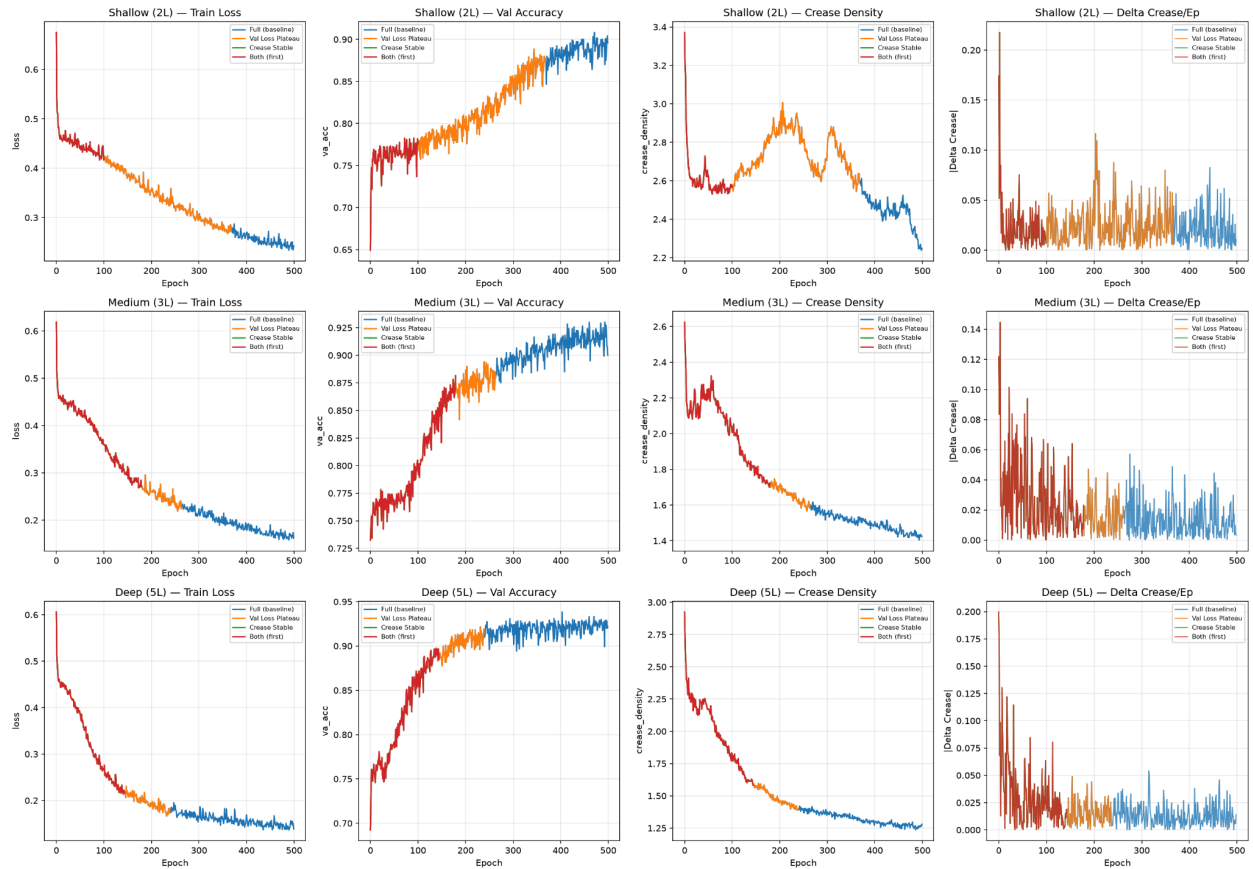




Ring Dataset (2D binary classification)







****Everything folds. Everything unfolds. The crease is where the action is.****

